Lab 4 — Predict Through Time

Friday, November 6 (~20 min)

Last updated: October 29, 2026.

In class Friday, November 6 (same day as Lecture 26 / Quiz 9). Quiz 9 is Lecture 23 + Lab 3 (products, powers, cost) — **not** this lab and not today's determinants.

Credit: Lab points are for effort in class, not for a Canvas file upload. Preview the one-pager; write on the hard copy handed out in class.

Cliffhanger for later. After Midterm 2, Lecture 35–37 + Lab 5 explain today's long-run mix with **eigenvectors** ($\lambda = 1$ direction wins).

Learning objectives

1. Implement one step of $x_{t+1} = Ax_t$.
2. Iterate several steps and hand-check the first ones.
3. Describe a long-run pattern you observe (you will explain it with eigenpairs in Lab 5).

Math connections

Linear dynamical systems (Lecture 22 and Lecture 23). Powers $A^k x$ appear in Lecture 23 and Lab 3.

Key syntax (this lab)

Task	Octave	Python
One step	`x = A*x`	`x = A @ x`
Loop	`for t = 1:20 … end`	`for t in range(20):`
Plot component	`plot(0:N, X(1,:))`	`plt.plot(range(N+1), X[0])`

Ops strip

On-/off-campus counts $x_t = \begin{bmatrix} c_t \\ o_t \end{bmatrix}$. Each week:

$$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}.$$

C4.0. Enter A and $x_0 = (1000, 0)$. Print $A * x_0$ / `A @ x0` (that is x_1).

Expected: $x_1 = (700, 300)$.

Application

C4.1. By hand, compute $x_2 = Ax_1$. Then check in software.

Expected: $x_2 = (550, 450)$.

C4.2. Starting from the same x_0, iterate 20 steps (store or print x_5 and x_{20}). Roughly where do the two populations settle?

Expected: toward about $(400, 600)$ (more off campus than on).

C4.3. Restart from $x_0 = (0, 1000)$. After 20 steps, what happens? In one sentence, what is surprising if you only look at the starting point?

Expected: again near $(400, 600)$. Different starts can head toward the same long-run mix — Lab 5 will name the missing idea.

One-pager

Preview the Lab 4 ticket; write on the hard copy in class.